

CS3391 - Object Oriented Programming

Topic: friend function

1. SUMMARY

A friend function is a non-member function that can access the private and protected members of a class. It is a special type of function that is declared as a friend in the class declaration. The concept of friend function is used to break the encapsulation of a class by allowing external functions to access its private and protected members. In Java, friend functions are not supported, but in C++, friend functions are used to give access to the private and protected members of a class.

The key principles of friend functions include:

- * Friend functions are non-member functions that can access the private and protected members of a class.
- * Friend functions are declared as friend in the class declaration.
- * Friend functions can access the private and protected members of a class.
- * Friend functions are used to break the encapsulation of a class.

Types and classifications of friend functions include:

- * Non-member friend functions
- * Member friend functions

Applications of friend functions include:

- * To break the encapsulation of a class by allowing external functions to access its private and protected members.
- * To provide a way to access the private and protected members of a class from outside the class.

Important points highlighted in the reference material include:

- * Friend functions are non-member functions that can access the private and protected members of a class.
- * Friend functions are declared as friend in the class declaration.
- * Friend functions can access the private and protected members of a class.
- * Friend functions are used to break the encapsulation of a class.

2. SHORT NOTES

****Detailed Definition with Examples****

A friend function is a non-member function that can access the private and protected members of a class. It is a special type of function that is declared as a friend in the class declaration. The concept of friend function is used to break the encapsulation of a class by allowing external functions to access its private and protected members.

For example, in the following code:

```
```cpp
class A {
private:
 int x;
public:
 friend void print(A a);
};

void print(A a) {
 cout << a.x << endl;
}
```
```

The function ``print`` is a friend function of class ``A`` and can access the private member ``x`` of class ``A``.

****Types and Classifications****

There are two types of friend functions:

- * ****Non-member friend functions****: These are friend functions that are not members of the class.
- * ****Member friend functions****: These are friend functions that are members of the class.

****Step by Step Working Principle****

The working principle of a friend function is as follows:

1. A friend function is declared as friend in the class declaration.
2. The friend function is a non-member function that can access the private and protected members of the class.
3. The friend function can be used to break the encapsulation of the class by allowing external functions to access its private and protected members.

****Important Formulas and Equations****

There are no specific formulas and equations for friend functions.

****Diagrams Description****

There are no specific diagrams for friend functions.

****Advantages and Disadvantages****

Advantages:

- * Friend functions can be used to break the encapsulation of a class by allowing external functions to access its private and protected members.
- * Friend functions can be used to provide a way to access the private and protected members of a class from outside the class.

Disadvantages:

- * Friend functions can compromise the encapsulation of a class.
- * Friend functions can make the class more complex.

****Real World Applications****

Friend functions are used in real-world applications to break the encapsulation of a class by allowing external functions to access its private and protected members. For example, in a banking system, a friend function can be used to access the private and protected members of a class that represents a bank account.

****Comparison with Related Concepts****

Friend functions can be compared with the following related concepts:

- * ****Member functions****: Member functions are functions that are members of a class and can access the private and protected members of the class.
- * ****Non-member functions****: Non-member functions are functions that are not members of a class and cannot access the private and protected members of the class.

****Time and Space Complexity****

The time and space complexity of a friend function depends on the specific implementation of the function.

****All Important Terminology****

The following are some important terminology related to friend functions:

- * **Friend function**: A friend function is a non-member function that can access the private and protected members of a class.
- * **Class**: A class is a blueprint or template that defines the properties and behavior of an object.
- * **Private member**: A private member is a member of a class that can only be accessed from within the class itself.
- * **Protected member**: A protected member is a member of a class that can be accessed from within the class itself and from within derived classes.

3. CODING EXAMPLES

C Implementation:

```
```c
#include <stdio.h>

class A {
private:
 int x;
public:
 friend void print(A a);
 void setX(int x) {
 this->x = x;
 }
};

void print(A a) {
```

```
 printf("%d\n", a.x);
}
```

```
int main() {
 A a;
 a.setX(10);
 print(a);
 return 0;
}
...
```

### Python Implementation:

```
```python  
class A:  
    def __init__(self):  
        self.__x = 0  
  
    def set_x(self, x):  
        self.__x = x  
  
    def print_x(self):  
        print(self.__x)  
  
def print_x(a):  
    print(a.__x)  
  
a = A()  
a.set_x(10)  
print_x(a)  
...
```

Algorithm

1. Create a class `A` with a private member `x`.
2. Declare a function `print` as a friend of class `A`.
3. Use the function `print` to access the private member `x` of class `A`.

Time and Space Complexity

- * Best Case: $O(1)$
- * Average Case: $O(1)$
- * Worst Case: $O(1)$
- * Space Complexity: $O(1)$

4. IMPORTANT QUESTIONS

Q1. (2 marks) What is a friend function in C++?

Answer: A friend function is a non-member function that can access the private and protected members of a class.

Q2. (2 marks) What is the purpose of a friend function?

Answer: The purpose of a friend function is to break the encapsulation of a class by allowing external functions to access its private and protected members.

Q3. (13 marks) Explain the concept of friend function with an example.

Answer: A friend function is a non-member function that can access the private and protected members of a

class. For example, in the following code:

```
```cpp
class A {
private:
 int x;
public:
 friend void print(A a);
};

void print(A a) {
 cout << a.x << endl;
}
```
```

The function `print` is a friend function of class `A` and can access the private member `x` of class `A`.

Q4. (15 marks) Explain the advantages and disadvantages of using friend functions.

Answer: Advantages:

- * Friend functions can be used to break the encapsulation of a class by allowing external functions to access its private and protected members.
- * Friend functions can be used to provide a way to access the private and protected members of a class from outside the class.

Disadvantages:

- * Friend functions can compromise the encapsulation of a class.
- * Friend functions can make the class more complex.

Q5. (2 marks) What is the difference between a friend function and a member function?

Answer: A friend function is a non-member function that can access the private and protected members of a class, while a member function is a function that is a member of a class and can access the private and protected members of the class.

Q6. (13 marks) Explain the time and space complexity of a friend function.

Answer: The time and space complexity of a friend function depends on the specific implementation of the function.

Q7. (15 marks) What are the types of friend functions?

Answer: There are two types of friend functions:

- * **Non-member friend functions**: These are friend functions that are not members of the class.
- * **Member friend functions**: These are friend functions that are members of the class.

Q8. (2 marks) What is the purpose of a friend declaration in a class?

Answer: A friend declaration is used to declare a function as a friend of a class, allowing it to access the private and protected members of the class.

Q9. (13 marks) Explain the concept of encapsulation in relation to friend functions.

Answer: Encapsulation is the concept of hiding the implementation details of a class and showing only the functionality to the user. Friend functions can compromise encapsulation by allowing external functions to access the private and protected members of a class.

Q10. (15 marks) Explain the difference between a friend function and a non-member function.

Answer: A friend function is a non-member function that can access the private and protected members of a class, while a non-member function is a function that is not a member of a class and cannot access the private and protected members of the class.

Q11. (2 marks) What is the purpose of a private member in a class?

Answer: The purpose of a private member is to hide the implementation details of a class and show only the functionality to the user.

Q12. (13 marks) Explain the concept of protected members in relation to friend functions.

Answer: Protected members are members of a class that can be accessed from within the class itself and from within derived classes. Friend functions can access protected members, allowing external functions to access the protected members of a class.

Q13. (15 marks) Explain the concept of inheritance in relation to friend functions.

Answer: Inheritance is the concept of creating a new class from an existing class. Friend functions can be used to access the private and protected members of a base class from a derived class.

Q14. (2 marks) What is the purpose of an interface in relation to friend functions?

Answer: An interface is a blueprint or template that defines the properties and behavior of an object. Friend functions can be used to access the private and protected members of a class that implements an interface.

Q15. (13 marks) Explain the concept of polymorphism in relation to friend functions.

Answer: Polymorphism is the concept of creating multiple classes that share a common base class. Friend functions can be used to access the private and protected members of a base class from a derived class.

5. MCQs

Q1. What is a friend function in C++?

- a) A member function that can access the private and protected members of a class.
- b) A non-member function that can access the private and protected members of a class.
- c) A function that is a member of a class and cannot access the private and protected members of the class.
- d) A function that is not a member of a class and can access the private and protected members of the class.

Answer: b) A non-member function that can access the private and protected members of a class.

Q2. What is the purpose of a friend declaration in a class?

- a) To declare a function as a member of a class.
- b) To declare a function as a friend of a class.
- c) To hide the implementation details of a class.
- d) To show only the functionality to the user.

Answer: b) To declare a function as a friend of a class.

Q3. What is the difference between a friend function and a non-member function?

- a) A friend function is a member function that can access the private and protected members of a class, while a non-member function is a function that is not a member of a class and cannot access the private and protected members of the class.
- b) A friend function is a non-member function that can access the private and protected members of a class, while a non-member function is a function that is not a member of a class and cannot access the private and protected members of the class.
- c) A friend function is a function that is a member of a class and can access the private and protected members of the class, while a non-member function is a function that is not a member of a class and can access the private and protected members of the class.

d) A friend function is a function that is not a member of a class and cannot access the private and protected members of a class, while a non-member function is a function that is a member of a class and can access the private and protected members of the class.

Answer: b) A friend function is a non-member function that can access the private and protected members of a class, while a non-member function is a function that is not a member of a class and cannot access the private and protected members of the class.

Q4. What is the purpose of a private member in a class?

- a) To hide the implementation details of a class.
- b) To show only the functionality to the user.
- c) To access the private and protected members of a class from outside the class.
- d) To access the public members of a class from outside the class.

Answer: a) To hide the implementation details of a class.

Q5. What is the difference between a protected member and a private member?

- a) A protected member is a member of a class that can be accessed from within the class itself and from within derived classes, while a private member is a member of a class that can only be accessed from within the class itself.
- b) A protected member is a member of a class that can only be accessed from within the class itself and from within derived classes, while a private member is a member of a class that can be accessed from within the class itself and from within derived classes.
- c) A protected member is a member of a class that can be accessed from within the class itself and from within derived classes, while a private member is a member of a class that can be accessed from within the class itself.
- d) A protected member is a member of a class that can only be accessed from within the class itself, while a private member is a member of a class that can be accessed from within the class itself and from within derived classes.

Answer: a) A protected member is a member of a class that can be accessed from within the class itself and from within derived classes, while a private member is a member of a class that can only be accessed from within the class itself.

Q6. What is the purpose of a friend declaration in a class?

- a) To declare a function as a member of a class.
- b) To declare a function as a friend of a class.
- c) To hide the implementation details of a class.
- d) To show only the functionality to the user.

Answer: b) To declare a function as a friend of a class.

Q7. What is the difference between a friend function and a non-member function?

- a) A friend function is a member function that can access the private and protected members of a class, while a non-member function is a function that is not a member of a class and cannot access the private and protected members of the class.
- b) A friend function is a non-member function that can access the private and protected members of a class, while a non-member function is a function that is not a member of a class and cannot access the private and protected members of the class.
- c) A friend function is a function that is a member of a class and can access the private and protected members of the class, while a non-member function is a function that is not a member of a class and can access the private and protected members of the class.
- d) A friend function is a function that is not a member of a class and cannot access the private and protected members of a class, while a non-member function is a function that is a member of a class and can access the private and protected members of the class.

Answer: b) A friend function is a non-member function that can access the private and protected members of a class, while a non-member function is a function that is not a member of a class and cannot access the

private and protected members of the class.

Q8. What is the purpose of a private member in a class?

- a) To hide the implementation details of a class.
- b) To show only the functionality to the user.
- c) To access the private and protected members of a class from outside the class.
- d) To access the public members of a class from outside the class.

Answer: a) To hide the implementation details of a class.

Q9. What is the difference between a protected member and a private member?

- a) A protected member is a member of a class that can be accessed from within the class itself and from within derived classes, while a private member is a member of a class that can only be accessed from within the class itself.
- b) A protected member is a member of a class that can only be accessed from within the class itself and from within derived classes, while a private member is a member of a class that can be accessed from within the class itself and from within derived classes.
- c) A protected member is a member of a class that can be accessed from within the class itself and from within derived classes, while a private member is a member of a class that can be accessed from within the class itself.
- d) A protected member is a member of a class that can only be accessed from within the class itself, while a private member is a member of a class that can be accessed from within the class itself and from within derived classes.

Answer: a) A protected member is a member of a class that can be accessed from within the class itself and from within derived classes, while a private member is a member of a class that can only be accessed from within the class itself.

Q10. What is the purpose of a friend declaration in a class?

- a) To declare a function as a member of a class.
- b) To declare a function as a friend of a class.
- c) To hide the implementation details of a class.
- d) To show only the functionality to the user.

Answer: b) To declare a function as a friend of a class.